

Implementation of synthetic inertia emulation in the WECC Generic Model for Large-scale PV Plants

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Abstract—The accurate modeling of photovoltaic power plants is essential for maintaining safe system operation as renewable energy penetration increases. This paper presents a review of current modeling approaches and provides a detailed analysis of the *WECC Generic Model for Large-scale PV Plants*, specifically regarding active power and frequency regulation.

A significant challenge in modern grids is maintaining frequency stability, which is traditionally dependent on the inherent inertia of synchronous generators, a property naturally absent in PV systems. To bridge this gap, this study proposes and implements a control strategy for synthetic inertia emulation within the industry-standard WECC Generic Model.

The proposed model is tested through simulations on a single-node system based on the Central European system separation event of January 8, 2021. Results demonstrate that the integration of synthetic inertia into the PV model significantly enhances frequency stability during disturbance scenarios, offering a viable solution for low-inertia power systems.

Keywords - generic model, photovoltaic power plants, synthetic inertia, single-node model, frequency stability

I. INTRODUCTION

The number of solar photovoltaic power plants has dramatically increased in the last decades in a global level as well as in Europe. Their contribution to the electricity supply reaches such a high level that in cases of appropriate conditions, it can exceed the other generation methods in terms of actual active power supply. [1]

Besides its numerous advantages, with such a large contribution of solar PV plants serious challenges have arisen. The most significant ones are the increased system losses, the worsening voltage and frequency stability [2] [3]. This paper aims to address the latter issue.

In order to thoroughly understand the impact on frequency stability of solar photovoltaic plants their modeling are inevitable and crucial. This can be performed by two different ways. In the first method the actual photovoltaic system is modeled in detail and provides a representation of the whole equipment. The drawback of these models is that they are manufacturer and hardware specific black-box models, whose implementation into distinct simulation software is challenging. The second method, called *generic* model is a standardized, non-proprietary white-box model.

A. Modeling techniques of solar photovoltaic power plants, a brief review

For the purpose of modeling the photovoltaic plants effect on frequency stability, a generic model which is capable of representing the behaviour of a real life PV plant independently from the manufacturer of the actual PV system was necessary [4].

Further advantages of a generic model is its *white box* structure, which enables other developers to perform modifications on the actual generic model. In addition, a generic model is widely compatible with the most common simulation software [5].

One of the most standard and widespread generic PV plant model is being developed by the Western Electricity and Coordinating Group (WECC), which is available in two versions; *WECC Generic Model for Distributed and Small PV Plants* and *WECC Generic Model for Large-scale PV Plants* [5]. This paper aims to deal with the latter one, which was previously validated and implemented in DlgSILENT PowerFactory [6].

B. Inertia in power systems

When there are no changes in the system, the difference of the electric and mechanical torque of a synchronous generator is zero. However in case of power imbalance, this balance fails and a frequency deviation occurs (1) [7] [8],

$$J \frac{d\omega}{dt} = T_m(t) - T_e(t) \quad \left[\frac{\text{kg} \cdot \text{m}^2}{\text{s}} \right] \quad (1)$$

where T_m is the mechanical torque of the turbine, and T_e is the electrical torque of the generator and J is the moment of inertia.

$$2H \frac{d\omega}{dt} = T_m(t) - T_e(t) \quad [\text{p.u.}] \quad (2)$$

Multiplying Eq. (1) by $\frac{1}{2} \frac{\omega_{base}^2}{S_{base}}$, the equation (2) can be obtained, where T_e and T_m are now expressed in p.u. and H is the inertia constant in s.

Equation (2) is also expressible for power for small deviations (3) or in the Laplace domain (4).

$$2H \frac{d\omega}{dt} = P_m(t) - P_e(t) \quad (3)$$

$$2H\omega s = P_m(s) - P_e(s) \quad (4)$$

Eq. (4) describes the inertial behaviour of a synchronous generator in case of power imbalance. The aim of this paper is to emulate this behaviour for a photovoltaic power plant.

C. Emulation of synthetic inertia in photovoltaic power plants

As a result of the lack of rotating mass, a photovoltaic power plant is unable to inherently provide inertial response. However this response can be emulated by (i) proper control strategy and (ii) an energy storage solution and called 'emulated synthetic or virtual inertia' [8].

The control is based on Eq. (4), and can be implemented as shown in Figure 1 [8], [9].

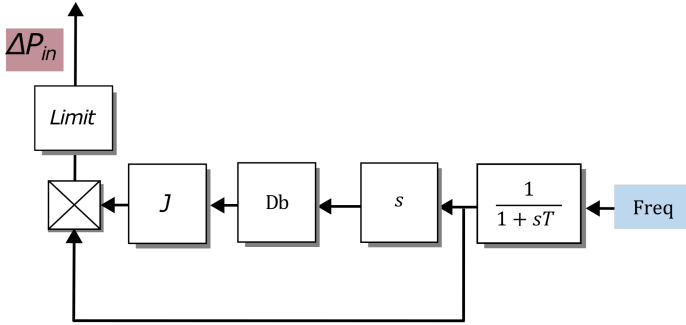


Fig. 1 — Synthetic inertia emulation in PV power plant [9]

In terms of energy storage, two main approaches exist, (i) the usage of ESS (batteries, super-capacitors, flywheels) and (ii) the 'de-loading technique'. This paper discusses the latter. In general, photovoltaic power plants operate at their maximal power point, according to the given weather conditions. Thus, in case of power imbalance, they are incapable of supporting additional power to the grid. However if a PV power plant operates below the maximum power point, it is capable to increase the output power and support the frequency control of the system [8]. The possible advantages of de-loaded operation of PV power plants were reported in [10].

II. THE IMPLEMENTATION OF SYNTHETIC INERTIA INTO WECC PHOTOVOLTAIC MODEL

A. The WECC Generic Model for Large-scale PV Plants

As mentioned in Section I-A, this paper deals with the WECC Generic Model for Large-scale PV Plants in DlgSILENT PowerFactory, implemented according to [6].

This model is a generic positive sequence renewable plant model. Among others it is suitable for studying the frequency response of different electrical disturbances, such as system separations. The model consists of three main control units: REPC_A, REEC_B and REGC_A as shown in Figure 2. A detailed description of these models has been previously reported in [5].

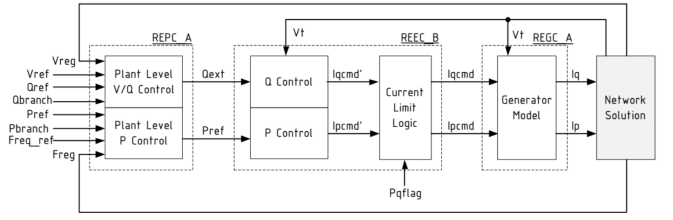


Fig. 2 — WECC Large scale PV plant model structure [5]

B. REPC_A active power control

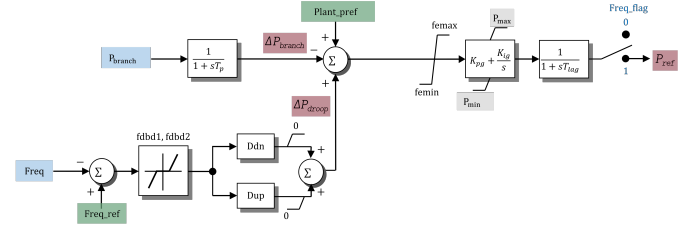


Fig. 3 — REPC_A active power control

Fig. 3 shows the active power control of the REPC_A control unit, which consist of a so called *droop control*. This control works according to Eq. (5) [9] and provides P_{droop} power control signal, which is proportional to the frequency deviation Δf . D_{droop} is the gain of the controller and fdb1, fdb2 each represent a distinct dead-band for the frequency deviation [5] [4].

$$\Delta P_{droop} = \Delta f \cdot D_{droop} = (f_n - f) \cdot D_{droop} \text{ [p.u.]} \quad (5)$$

C. Emulation of synthetic inertia

As it is shown in Fig 3 the active power control of the original WECC model lacks the emulation of the synthetic inertial response. To overcome this issue this paper presents a possible implementation of synthetic inertia in the WECC Generic Model for Large-scale PV Plants.

The synthetic inertia control is based on Eq. (4) and Fig. 1.

The signal ΔP_{in} is connected to the summation point of the REPC_A control unit, as shown in Fig. 4. ΔP_{in} is multiplied by -1 because even though if P_m is smaller than P_e additional power would be needed, the derivative of the frequency is negative, so the value of ΔP_{in} is also negative. This means that the photovoltaic power plant would decrease its active power output, leading to further frequency deviation. To prevent this behaviour the multiplication of ΔP_{in} by -1 is essential.

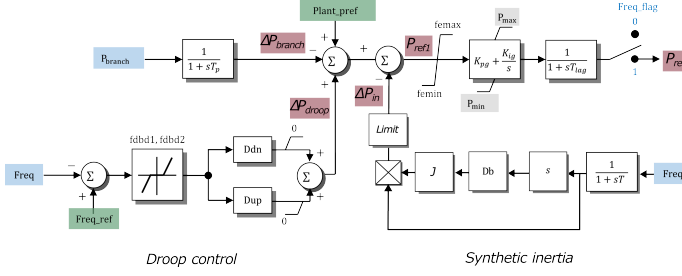


Fig. 4 — REPC_A active power control with emulated synthetic inertia

The supplemented *WECC Generic Model for Large-scale PV Plants* is now capable of emulating a synthetic inertial response for both over- and under-frequency scenarios when it is operated in de-loaded mode.

III. CASE STUDIES

A. The test configuration

The tests were based on a real-world event in which the power grid experienced both increasing and decreasing frequencies. For this, the European power system separation that occurred on January 8, 2021 was chosen. This event is well documented and has already been analyzed using several models reported in other studies [11, 12]. During this incident, the system separates into two regions, a *South-East* (from now on: EAST) and a *North-West* (from now on: WEST) part. In the EAST, significant overproduction occurred, causing the system frequency to increase. In contrast, the WEST region experienced underproduction, which led to a decrease in frequency.

To keep the model as simple and as computationally efficient as possible, it was opted for a single-node model, reported in [11], which accurately reproduces the real frequency response measured on 8 January 2021. In this paper the same dataset for the model was used as in [11].

However, our aim was to examine the impact of increased photovoltaic generation on the grid. Therefore, in the single-node model, the model of the system dynamics, which was originally represented by a synchronous generator [11], was supplemented with a *photovoltaic power plant*, which was based on the extended *WECC Generic Model for Large-scale PV Plants*. The resulting single-node model structure is shown in Fig. 5. The rest of the model was build up, and the parameters of the model was set up based on [11].

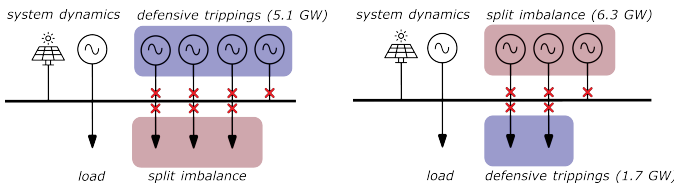


Fig. 5 — EAST and WEST single node models supplemented with PV production

The developed model was thoroughly tested in different grid scenarios and parameter settings. The simulations focused on

changes in the following parameters: (i) PV penetration ratio [%], (ii) value of synthetic inertia constant (J [p.u.]).

B. The effect of PV contribution

In this case, the examination focused on the effect of the photovoltaic penetration ratio; therefore, the value of the synthetic inertia constant was kept unchanged ($J = 25$).

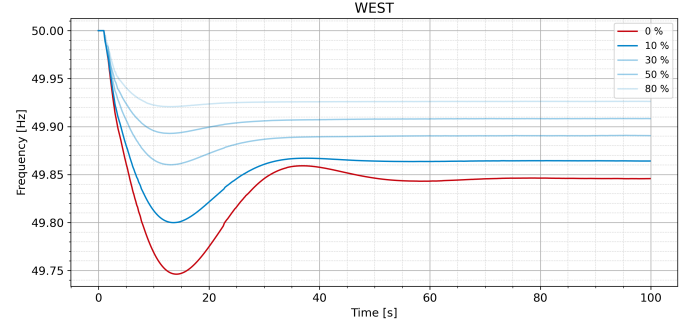


Fig. 6 — Frequency response WEST - $J = 25$

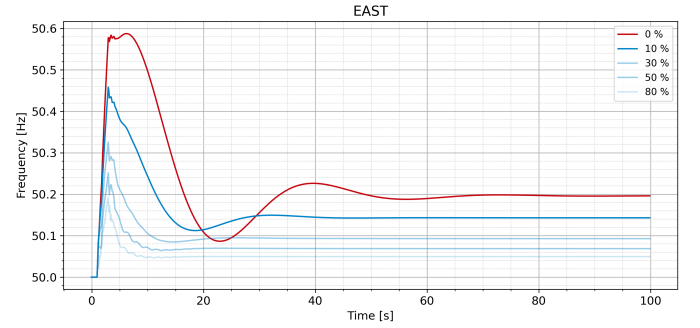


Fig. 7 — Frequency response EAST - $J = 25$

The simulations show that higher photovoltaic penetration leads to a more stable frequency response in all cases. This behavior can be explained by two main factors. First, PV systems can respond faster than synchronous generators due to the absence of mechanical inertia, provided that an appropriate control strategy is applied. Second, the PV units were operated in a de-loaded mode, which allowed them to increase their output during frequency drops, particularly in the WEST case where additional generation was required. It should be noted that without de-loaded operation, or an additional energy storage system, synthetic inertia alone cannot effectively support the frequency response.

C. The effect of inertia constant

In this case, the examination focused on the effect of the value of the synthetic inertia constant; therefore, the photovoltaic penetration ratio was kept unchanged ($PV_{pen} = 30\%$).

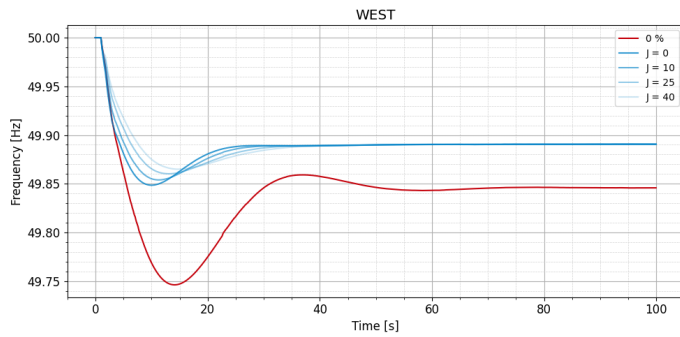


Fig. 8 — Frequency response WEST - 30% PV penetration

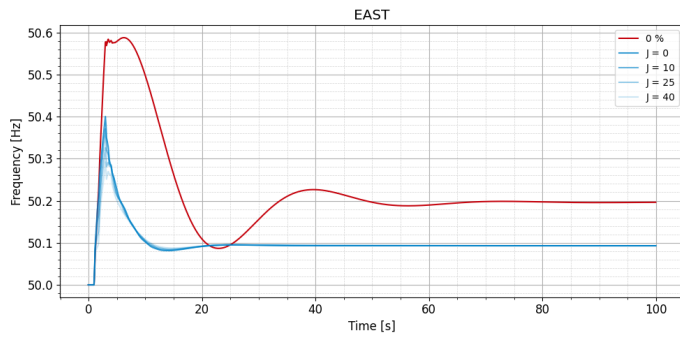


Fig. 9 — Frequency response EAST - 30% PV penetration

Increasing the inertia constant J results in an improved frequency response. This is because a higher inertia constant leads to a more aggressive synthetic inertia contribution from the photovoltaic system. As illustrated in Fig. 1, larger values of J correspond to stronger control intervention.

IV. CONCLUSION

This paper presented a possible implementation of synthetic inertia emulation in the industry standard *WECC Generic Model for Large-scale PV Plants*.

With the created model numerous tests were performed on a single-node model which based on the Central European system separation in January 8, 2021. The results of these tests clearly show the advantageous result of photovoltaic power plants and the emulated synthetic inertia in terms of frequency stability.

Further investigation could be conducted to fine-tune synthetic inertia emulation and validate its behaviour with real life scenarios [13].

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